Martin Wassim

Software Engineer | AI/ML Engineer | Founder

CONTACT

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SKILLS

Programming
Python, C#, Java, VB.NET, JavaScript, TypeScript, SQL, HTML5, CSS3

Backend & Web
ASP.NET Web API, REST APIs, Flask, Next.js, React, Tailwind CSS, Supabase, Full-Stack Development

AI & Machine Learning
PyTorch, TensorFlow, Deep Learning, Computer Vision, CNNs, Vision Transformers (ViT), NLP

Databases
SQL Server, MySQL, Relational Database Design

Desktop
VB.NET WinForms, Java Swing, Tkinter, MaterialSkin, Apache POI

Tools & Security
Git, Postman, RestSharp, Kali Linux, Nmap, Burp Suite

Other
OOP, Requirements Gathering, Product Lifecycle Management

CERTIFICATES

Artificial Intelligence (90 hrs)
Information Technology Institute (ITI)

HCIA-AI V4.0 Course
Huawei ICT Academy

Fundamentals of Deep Learning
NVIDIA

Getting Started with Deep Learning
NVIDIA

Deep Learning in Python
DataCamp

NLP in Python
DataCamp

Image Processing in Python
DataCamp

Computer Vision & Intro to Deep Learning
Kaggle

Python Essentials 1
Cisco Networking Academy & Python Institute

EDUCATION

B.Sc. Computer Science
Robotics and Software Engineering
Helwan National University
09/2023 – 09/2027

LANGUAGES

Arabic – Native
English – Upper-Intermediate

PROFILE

Computer Science student and software engineer with hands-on experience building full-stack, AI/ML, desktop, and enterprise applications. Founder of Systemize EG, delivering custom business software, CRM solutions, automation systems, and management tools. Experienced with Python, C#, Java, ASP.NET Web API, SQL Server, PyTorch, and TensorFlow, with additional depth in computer vision, deep learning, and software architecture.

EXPERIENCE

Founder
12/2025 – Present

Systemize EG · Egypt

- Designed and delivered a custom **Daily Sales Management System (POS)** with an Arabic-first UI, multi-payment support, and Excel reporting using Java Swing and Apache POI.
- Designed CRM solutions and smart automation systems for business clients.
- Developed websites and digital tools to help businesses work smarter and scale faster.
- Managed the full product lifecycle: requirements, development, delivery, and client support.

Software Development Intern
08/2026 – 09/2026

National Steel Fabrication (NSF) – Orascom Construction · Egypt

- Built an enterprise IT request and workflow system using VB.NET WinForms and a responsive web dashboard.
- Developed a C# ASP.NET Web API integrated with SQL Server for request routing, status management, and audit tracking.
- Created a responsive management dashboard (JavaScript, HTML5, CSS3) for employee requests and ticket monitoring.
- Designed a normalized SQL Server database with transactional integrity and sequential ticket ID generation.
- Implemented requester workflows for personal and “On behalf of” submissions with dynamic manager lookup.

PROJECTS

Velora – Daily Mission Game

Next.js 16 · React 19 · TypeScript · Tailwind CSS · Supabase (Auth, Postgres, Storage) · Vercel

- Built a full-stack daily mission game with XP, coins, levels, streaks, achievements, daily login rewards, and a leaderboard.
- Developed an admin dashboard for user and role management, mission creation and assignment, submission review (approve/reject with bonus XP and coins), rewards, and activity logs.
- Enforced rewards server-side with server actions and Postgres security rules, so players cannot modify their own XP or coins.
- Implemented photo-proof uploads, automated 24-hour mission publishing via a Vercel cron job, and phone home-screen install (PWA).

WasteNet – Waste Classification Neural Network

Python · PyTorch · Flask · CNN · Vision Transformer

- Developed a deep learning system that classifies waste into six categories.
- Benchmarked baseline CNN, improved CNN, transfer learning, and Vision Transformer (ViT) architectures.
- Deployed the best model as a local inference server with an interactive web dashboard.

RAG-Powered Document Assistant

Python · RAG · LLM · ChromaDB · Sentence Transformers · Ollama · FastAPI · Streamlit

- Built a Retrieval-Augmented Generation system for question answering over academic PDF documents.
- Implemented document extraction, chunking, semantic search, and vector storage with Sentence Transformers and ChromaDB.
- Integrated a local LLM via Ollama to generate grounded, citation-backed answers based only on retrieved content.
- Delivered a FastAPI backend and Streamlit chat interface with automated testing and Docker-based deployment support.

Enterprise IT Request & Dashboard System (NSF – Orascom Intern Project)

C# · ASP.NET Web API · VB.NET · SQL Server · JavaScript

- Built an end-to-end enterprise ticketing platform with desktop and web interfaces.
- Developed REST APIs for ticket routing, status transitions, and creator audit tracking.
- Designed the SQL Server relational database and responsive web dashboard for monitoring and administration.

Coin Counting & Classification System

Python · OpenCV · Computer Vision · Image Segmentation

- Developed a computer vision system for automated coin detection, counting, classification, and value calculation.
- Applied preprocessing, Otsu thresholding, morphological operations, distance transform, and watershed segmentation.
- Compared thresholding techniques and evaluated performance using accuracy, MAE, and RMSE.

VRP Solver – Genetic Algorithm & Differential Evolution

Python · Tkinter · Evolutionary Algorithms

- Built a desktop application comparing Genetic Algorithms and Differential Evolution for capacity-constrained vehicle routing, with interactive route visualization.

Hospital & Healthcare Website

HTML · CSS · JavaScript

- Developed a responsive healthcare platform with appointment booking, doctor search, telemedicine, laboratory booking, patient portal, and administration features.

Information System & Network Security Pentest Project

Kali Linux · Nmap · Burp Suite · Web Security

- Penetration-tested a vulnerable e-commerce system, exploiting SQL Injection, XSS, Command Injection, and Privilege Escalation.
- Performed network enumeration, password cracking, and security analysis with full technical documentation.